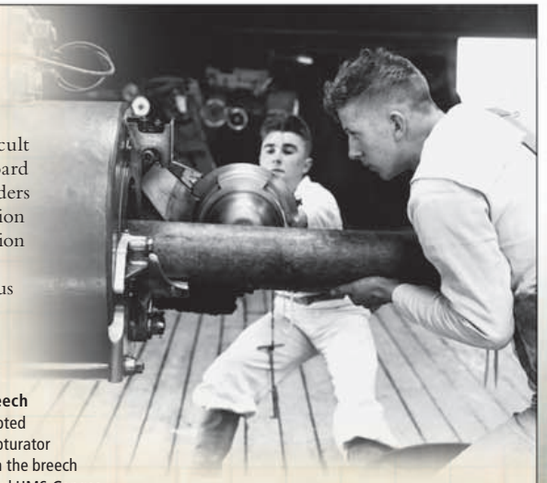
WEAPONS AND TECHNOLOGY

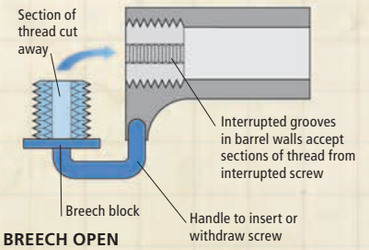
BREECH-LOADING GUNS

In the mid-19th century the breech-loading rifled cannon began to replace the muzzle-loader, which took longer to load, was difficult to clean, and could mistakenly be double-loaded, as happened aboard HMS *Thunderer* in 1879, causing terrible loss of life. But breech-loaders had dangers of their own, and only became reliable with the invention of the interrupted screw and the French-designed De Bange obturation system. The former allowed the breech to be opened and closed quickly, while the latter created an airtight seal, preventing dangerous propellent gasses escaping through the breech when firing.

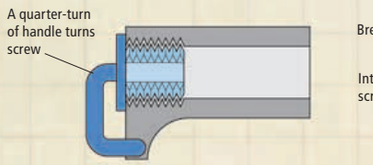


Sealing the breech
Both the interrupted screw and the obturator pad are visible in the breech of this gun aboard HMS *Ganges*.

Interrupted screw
This features a screw thread that has sections cut away lengthwise. This allows the breech block to be fully inserted into the barrel before being quarter-turned shut.

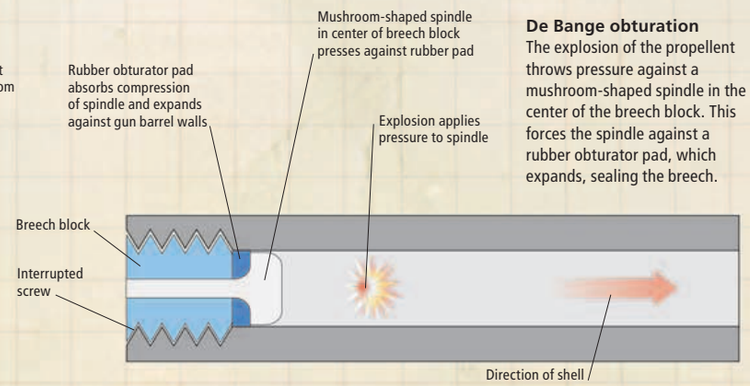


BREECH OPEN



BREECH CLOSED

De Bange obturation
The explosion of the propellent throws pressure against a mushroom-shaped spindle in the center of the breech block. This forces the spindle against a rubber obturator pad, which expands, sealing the breech.



SPANISH-AMERICAN WAR



THE BATTLE OF MANILA BAY

Date May 1,1898
Location Off Cavite, Philippines
Result American victory

COMBATANTS

UNITED STATES	SPAIN

COMMANDERS

George Dewey	Patricio Montojo y Pasaron

FORCES

Ships: 6 warships, 3 support vessels	**Ships**: 7 warships

LOSSES

Men: 9 minor casualties **Ships**: none	**Men**: 381 men killed or wounded **Ships**: 7 ships lost

